

Homework 1 — Big-O Analysis and Arrays

Exercise 1. For each of the following functions, give the tightest Big-O bound and justify your answer in one or two sentences: (a) $3n^2 + 5n \log n$, (b) $2^n + n^{10}$, (c) $n \log n + 100n$, (d) $\sqrt{n} + n$.

Exercise 2. You insert n elements one at a time into a dynamic array that starts at capacity 1 and doubles when full. Show that the total cost of all n insertions is $O(n)$ using the accounting method.

Exercise 3. Implement a function that removes all duplicates from an unsorted array of integers in place. State the time and space complexity of your solution, and describe a faster approach that uses a hash set.

Exercise 4. Write pseudocode to rotate an array of n elements to the left by k positions using only $O(1)$ extra space. Your solution must run in $O(n)$ time and must not use a second array.

Submission rules: submit a single PDF containing your written answers and any code as a readable monospace block. Homework is due at the start of lecture in week 4; late submissions lose 10% per day.